



Iriscom's Quick Glance 2 follows your eyes and clicks on icons when you blink slowly

It's clear that enough inroads have been made in this field to already have viable products. As for "tomorrow," things can only get better and more accurate!

Read My Mind

Wouldn't it be great if PCs could read your mind? Well, they can! Back in March 2005, *Wired Magazine* (www.wired.com) wrote an article about Matthew Nagle, a victim of a violent crime that left him with a severed spine—a quadriplegic for life. Nagle, thanks to some innovative technology research, was able to regain a little control of his life. After his doctor suggested that he join a clinical test being conducted by Cyberkinetics Neurotechnology Systems (www.cyberkinetics-inc.com), Nagle received a brain implant. After intensive training, Nagle can now turn his TV set on and off, raise the volume, and change channels. He can also interface with his computer and play *Pong*, draw basic shapes, and more.

The chip in his brain connects to software in his PC or home appliances, and Matthew "wills" them to work by *thinking* about what he wants to do. Thanks to Cyberkinetics Neurotechnology Systems' BrainGate Neural Interface System, Nagle has hope. Though the clinical trials are only open to people suffering from similar condi-

tions as Nagle, the technology works! We now know that thought control is actually possible, and research will get us there soon enough!

A Brain Computer Interface was recently showcased, at CeBIT 2006. It allowed two people to put on some strange-looking headgear and then play *Pong*—perhaps one of the simplest games ever made. Yes, the headgear is funny, and the sheer number of wires emanating from it are likely to have people cracking "Please don't hurt me, I'll take you to my leader!" jokes. But the fact is, we haven't reached the stage of aesthetics yet. When (with emphasis on the word *when*) such technologies are perfected, we can start worrying about aesthetic appeal; for now, we're just happy that it works.

There are many research projects working on this new way of interfacing with computers, and that's heartening for those of us who are fed up of keyboards and mice! Here's Wikipedia's entry for Brain-Computer interface, which should help you find more interesting news about this exciting technology change: http://en.wikipedia.org/wiki/Brain-computer_interface.

We Don't Want Silly Hats

Well, we don't! We don't want to have to sit stationary in front of our computers either, as is required by head or eye tracking systems. The same holds true for the gloves and other devices that connect to limbs and allow you to control motion. Most of us would not be willing to undergo surgery for implants either, for fear of short-circuiting our brains! So what do we do?

As the research, prototypes and products mentioned here already suggest, researchers are already looking for a non-intrusive, minimal-fuss interface that will only limit you as much as a mouse and keyboard do.

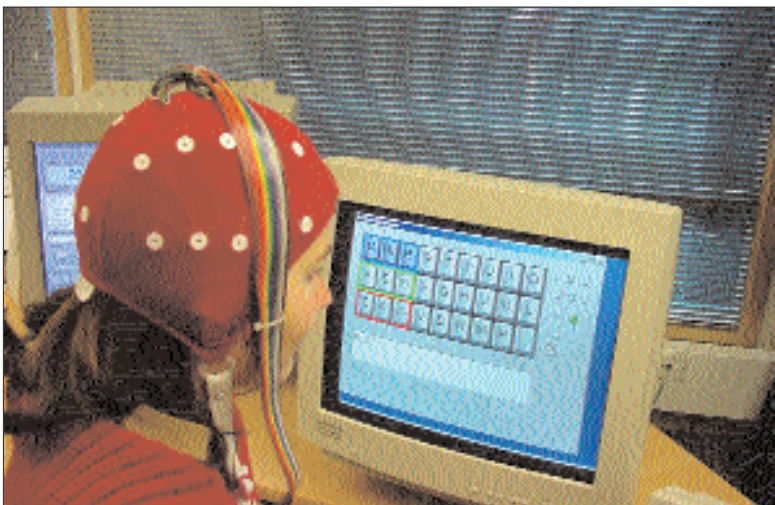
What Happens Tomorrow?

In the near future—five years or less—we hope to be able to see a coming together of all these technologies and methods of interfacing. This is the only way in which a truly complete solution might emerge.

Think about it: thought control will never be perfect, because human brains are unique enough to cause anomalies in the way the software registers thought patterns. What is "Minimise" for me might be "Close" for you and so on. What can help iron out the creases are speech and motion detection software. If you think "Minimise," and the PC is confused about whether you meant "Minimise" or "Close," it can use the eye tracking camera and software to figure out which button you were looking at. Or perhaps it can use a little AI to understand that your body language does not suggest that you're done with the window in front of you, and therefore would not want to close it.

If someone could integrate speech recognition (or natural language processing, preferably), motion detection, thought control and a little bit of AI, we might finally be rid of our traditional, boring input devices. Until that day, nothing is going to dethrone them—old habits die hard! ■

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The cap looks silly, but typing with your brain is cool! Here's another Brain Computer Interface under development at the Laboratory of Computational Engineering in Finland